For all calculations – <u>show your work including equations used</u>. You do not need to show units in the calculations but make sure to show your final answer in the correct units (and units do provide a built-in check).

Q1 (5 points)

A sample of unknown gas A (g) escapes through a tiny hole into a vacuum chamber in 187.5 s. The same concentration of unknown gas X_2 (g) escapes through the hole into the evacuated vacuum chamber in 101.3 s under identical conditions. You are given that X_2 (g) is a homonuclear diatomic gas, and that a 13.129 g sample of A (g) in a 1.500 L flask at 25.00 °C has a measured pressure of 1.631 atm. Identify both A (g) and X_2 (g). Justify your answer with calculations. (4 significant figures)

Q2 (10 points)

a) 7 points

The rate of the reaction of phosphine with diborane was measured as a function of temperature.

PH_3 (g) + B_2H_6 (g) → H_3P-BH_3 (g) + BH_3 (g)

A plot of ln(k) versus the reciprocal of absolute temperature obtained from the kinetic data for this reaction had a slope of -5773 K.

At what temperature (in K) would you predict a doubling of the rate for this reaction relative to the rate at room temperature 25.00 ^{o}C? Justify your answer showing full calculations. (4 significant figures)

b) 3 points

There is a rough “rule of thumb” in chemical kinetics that predicts a doubling of the rate of a reaction for every 10 degrees increase in temperature.

Does this approximate kinetic rule of thumb estimate hold for the reaction in part a)? Briefly provide a physical chemical justification in terms of gas molecular behavior for this approximate kinetic temperature relationship.

Q3 (10 points)

A solution of a newly discovered polymer is prepared by dissolving 30.00 milligrams of polymer in 1.00 millilitre of water. This sample exerts an osmotic pressure of 1239.5 Pascal at 25˚C.

What is the molecular weight of the polymer? (3 significant figures)

Q4) 10 points

A new element has been discovered, Sewallium (Sw)! It forms an ionic compound with chlorine that acts as a strong electrolyte in water. You are in charge of determining the number of chloride ions bound to each Sewallium ion, using freezing point depression.
In a carefully controlled experiment a 1.00×10^{-3} molal solution of Sewallium Chloride in water has a freezing point depression of 0.00735 K.

Using this measurement determine the most likely chemical formula for Sewallium Chloride from among the following options: $SwCl$, $SwCl_2$, $SwCl_3$, $SwCl_4$. Justify your answer showing relevant calculations.

Q5) 6 points

A sample is prepared by adding 7271.7 grams of ethylene glycol to 69308.9 grams of water, which forms a solution that has a total volume of 75.68 liters.

The molecular weight of ethylene glycol is 62.07 gr/mol.
The molecular weight of water is 18.02 gr/mol.

Calculate the concentration of ethylene glycol in this solution expressed as (3 significant figures for each):
a. mole fraction

b. molality

c. molarity

Q6) 10 points

Consider the following reaction, where $K_c = 1.57*10^{-26}$:

$$2\,A(g) + 3\,B(g) \longrightarrow C(g) + D(g)$$

Calculate the equilibrium concentrations of A, B, C, D starting with 0.30 mol A and 0.60 mol of B combined in total volume 1.00 L. Note that none of the concentrations can be zero at equilibrium. (3 significant figures)

Q7) 5 points

For the reaction

$$A(g) \rightleftharpoons B(g)$$

$K = 1.87 * 10^{-7}$
$\Delta S^0 = 181.92$ J/(mol*K)

a. What is the value of ΔH^0 at 298 K? (3 significant figures)

b. At what temperature would a mixture containing an equal number of moles of A and B be at equilibrium? (3 significant figures)

Q8) 12 points

a) 4 points

K_a of HX is 4.65 x 10^{-5}. What is the pH of a 0.500 M solution of NaX at 298 K? (3 significant figures)

b) 8 points

Arrange the following from lowest to highest pH, using the information given in the table. Anions are bolded. Explain your answer.

0.1 M HB**Cl**; 0.1 M HA**X**; 0.1 M HB**X**

Ion or Molecule	**pK_a**	**pK_b**
AH^+	8.61	
HX	5.39	
B		3.20

Q9) 15 points

a) 4 points

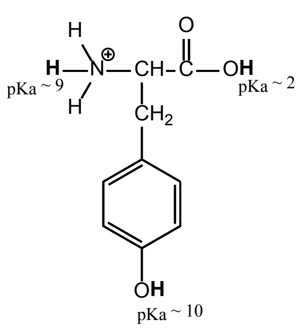
In a chemistry experiment, you must choose possible reagents to carry out a reaction to form the highly reactive anionic base, X^-, from it's conjugate acid (pK_a of the acid, HX, is 25).

Using the data below, identify the reagent(s) you could use to carry out the reaction? You can choose more than one reagent. You must provide the reagent(s) in its correct form (acid or the conjugate base) and justify your answer. You do not need to provide counter-cation or anion.

Acid	**pK_a**
HA	10
HB	20
HC	35
HD	50

b) 6 points

Draw the most prominent ionization state for the following compound at (i) pH = 7.50 and (ii) pH = 12.00, at 298 K using the data given below. The pK_a values given are for the hydrogens shown in bold. Justify your answer.



c) 5 points

K_a for HX is 2.5 x 10^{-8}, calculate the equilibrium constant (*K)* value for the reaction of $HX_{(aq)}$ with $OH^{-}{}_{(aq)}$ at 298 K. Show relevant equations and calculations. (3 significant figures)

Q10) 6 points

For a polyprotic acid, H_2A:

$K_{a,1} = 1.50 \times 10^{-3}$

$K_{a,2} = 2.50 \times 10^{-11}$

You need to conduct an important kinetic assay at pH = 10.5. Using the data above, choose a buffer system (acid/conjugate base) that can be used for your assay. Justify your answer.

What is the ratio of acid to conjugate base at pH =10.5? Show calculations.

Q11) 5 points

In one of their missions for the starship Enterprise, Spock and Sulu discover a new element, A. They calculate the solubility of the ionic salt, ACl_3, in water to be 5.87 mg/L. What is the K_{sp} of ACl_3? Molecular weight of ACl_3 is 393 g/mol. Show your work and calculations. (3 significant figures)

Q12) 16 points

20.00 mL of a 0.1000 M solution of a weak base (K_b = 2.2 x 10^{-5}) is titrated with 0.1500 M strong acid, HCl, at 298K.

i. Calculate the initial pH of the solution

ii. Calculate the pH of the solution after addition of 5.00 mL of HCl. Show your calculations (3 significant figures)

iii. Calculate the pH of the solution after addition of 30.00 mL of HCl. Show your calculations (3 significant figures)

iv. Which of the following indicator(s) would you choose for this titration? Justify your choice with calculations.

Indicator	**pK_a(s)**	**Indicator**	**pK_a(s)**
Thymol Blue	1.65 and 8.9	Methyl Yellow	3.3
Bromophenol Blue	3.85	Methyl Red	4.95
Phenolphthalein	9.4	Alizarin Yellow	11.2

Q13) 10 points

a) 4 points

Balance the following equation using smallest whole number coefficients. Show half-reactions and the final fully balanced equation using smallest whole number coefficients.

In basic media

$$X_{2(aq)} \rightleftharpoons X^{-}(aq) + XO_4^{-}(aq) + H_2O(l)$$

(X is a non-metal and is less electronegative than O)

b) 6 points

A voltaic cell is constructed using the reagents based on the following half reactions:

$$X^{2+}(aq) + 2e^- \rightleftharpoons X(s) \qquad E^o = -0.22$$

$$Y^{3+}(aq) + e^- \rightleftharpoons Y^{2+}(aq) \qquad E^o = -0.51$$

Calculate the initial voltage (2 significant figures) generated by the cell at 25 ˚C, if the initial concentration of X^{2+} (aq) = 0.20 M; Y^{2+} (aq) = 0.25 M; and Y^{3+} (aq) is 0.0020 M.

Show your work, including any equations used, and calculations.

Q14) 12 points

a. 9 points

Some batteries consist of the Zn (s) and MnO_2 (s) electrodes. At the Zn electrode ZnO(s) forms, and at the MnO_2 electrode Mn_2O_3(s) forms.

Using the above information, write the half reactions for both anode and cathode, and the balanced overall reaction. Show your work.

Calculate the mass in kg (3 significant figures) of ZnO (s) formed if 1.0×10^4 A are passed through the cell for 10 hours. Show your work.

b. 3 points

What are the electrolysis products for the following mixture? Justify your answer.
$AlCl_3$ + KBr (molten)

Q15) 12 points

Using the table given below, answer the following questions:

Half reactions	E° (volts)
$A^+ (aq) + e^- \rightarrow A(s)$	1.73
$B^{2+} (aq) + 2e^- \rightarrow B(s)$	-0.14
$C^{2+} (aq) + 2e^- \rightarrow C(s)$	-0.41
$X_2 (g) + 2e^- \rightarrow 2X^- (aq)$	1.28
$D^{2+} (aq) + 2e^- \rightarrow D(s)$	-0.17
$E^{2+} (aq) + e^- \rightarrow E^+(aq)$	-0.29

a) 2 points

Which reagent is the strongest reducing agent? Justify your choice briefly.

b) 4 points

Which reagent(s) will reduce B^{2+} (aq) while leaving C^{2+} unreacted? Justify your choice briefly.

c) 6 points

You want to obtain a voltaic cell that has the highest cell potential from the data given in the table. Write down the overall balanced equation and the E° cell value for the cell with the highest potential. Justify your choice.